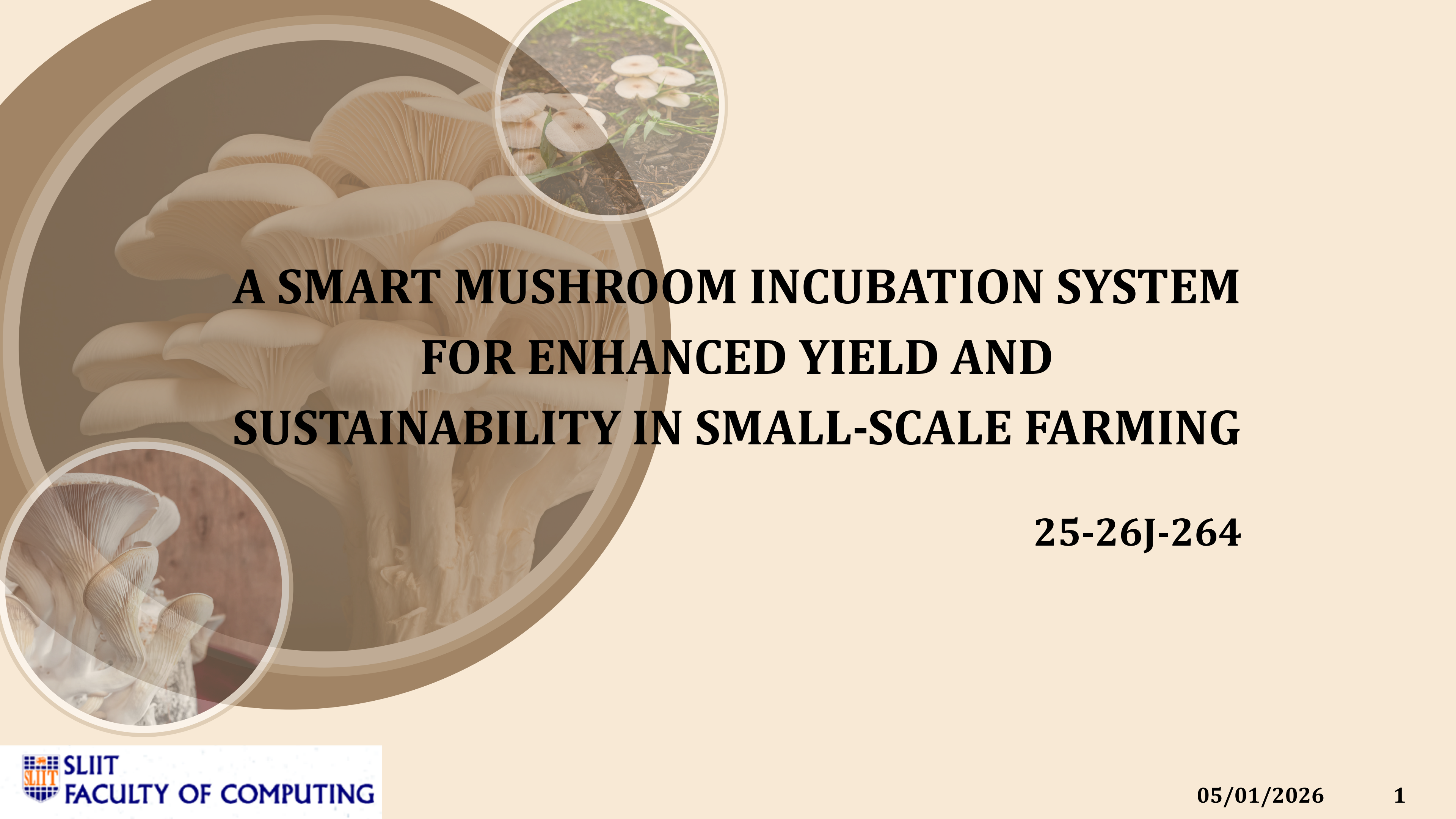




A SMART MUSHROOM INCUBATION SYSTEM FOR ENHANCED YIELD AND SUSTAINABILITY IN SMALL-SCALE FARMING

25-26J-264



LEARN ABOUT

« A SMART MUSH ROOM
INCUBATION SYSTEM FOR
ENHANCED YIELD AND
SUSTAINABILITY IN SMALL-SCALE
FARMING. »

IT22063182 | M L Vidanaarachchi

Predictive Fruiting Module using Environmental Sensing and Machine Learning

Specialization: Information Technology



Predictive Fruiting Module using Environmental Sensing and Machine Learning

How the Solution Addresses the Sub-Problem

- Continuously collects environmental data (temperature, humidity, CO₂) using sensors
- Real-time sensor data is stored and synchronized via Firebase

Prototype Status

- Robot arm and sensor setup completed
- Real-time sensor data successfully displayed in database and mobile application
- Mushroom detection and condition classification models trained and validated

User Requirements Addressed

- Real-time visibility of farm environmental conditions
- Reduced need for manual monitoring
- Simple mobile interface for non-technical users

Predictive Fruiting Module using Environmental Sensing and Machine Learning

Design Excellence & Contribution

- Modular design allows environmental sensing, vision models, and prediction logic to operate independently
- Integration with Firebase ensures real-time data flow between sensors, AI models, and the mobile app

User Feedback on Prototype

- Real-time sensor data visualization improves farmer awareness of growing conditions
- Mobile app interface is easy to understand and reduces manual farm checks

Next Phase Enhancement

- Train and integrate a time-series ML model to predict fruiting readiness based on sensor data
- Provide estimated fruiting time with confidence levels through mobile alerts

IT22586384 | S W O P K Waduge



Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

Specialization: Information Technology

Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

Predictive Advisory System – Solution & Prototype (Current Stage)

- IoT sensors collect real-time substrate moisture, pH, and environmental data
- Continuous data streaming and storage via Firebase
- Web and mobile dashboards display live substrate conditions

Prototype Status

- Sensor integration and real-time data pipeline completed
- Front-end dashboards functional
- Advisory logic and predictive model implementation pending

User Requirements Addressed

- Real-time visibility of substrate conditions
- Centralized monitoring through web and mobile interfaces
- Reduced manual measurement effort

Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

Design Excellence, Contribution & Preliminary Feedback

- Modular design separates data collection, storage, and advisory layers
- Real-time dashboards improve operational awareness
- Shared Firebase infrastructure ensures smooth system-wide integration

User Feedback

- Real-time visibility of substrate moisture and pH is useful for daily monitoring
- The dashboard simplifies understanding of changing substrate conditions
- Continuous data collection reduces manual measurement effort

Next Phase Enhancements

- Develop advisory engine with actionable recommendations and confidence levels
- Validate recommendations through real-user field testing

IT22290960 | B H A S Peiris



Fungal Contamination Detection Module using Image Processing and Deep Learning

Specialization: Information Technology

Fungal Contamination Detection Module using Image Processing and Deep Learning

Fungal Contamination Detection – Solution & Prototype

- Motorized camera performs row-by-row scanning of mushroom racks
- Images of mushroom packets captured and streamed to the system
- Deep learning model trained to identify mushrooms from images

Prototype Status

- Image capture mechanism implemented
- Mushroom identification model trained

User Requirements Addressed

- Early detection of fungal contamination
- Reduced manual visual inspection
- Faster response to prevent spread

Fungal Contamination Detection Module using Image Processing and Deep Learning

Design Excellence / Contribution

- Automated visual scanning reduces human error
- Modular vision pipeline supports future disease expansion

User Feedback (Internal Evaluation)

- Automated scanning improves inspection efficiency
- Visual detection preferred over manual checks
- Early contamination detection seen as highly valuable

Next Phase Enhancements

- Real-time inference and alert notifications
- Expansion of disease classes and dataset

IT22062888 | G A I W Wijesinghe



Pest Management Module using Real-Time Detection and Automated Response

Specialization: Information Technology

Pest Management Module using Real-Time Detection and Automated Response

Solution & Prototype

- The camera platform patrols mushroom racks
- Images captured continuously for pest monitoring
- Stage 1: Model trained to identify mushrooms
- Stage 2: CNN model trained to detect pests

Prototype Status

- Camera movement and image capture implemented
- Pest detection model trained

User Requirements Addressed

- Continuous pest monitoring
- Early pest identification
- Reduced chemical usage through targeted response

Pest Management Module using Real-Time Detection and Automated Response

Design Excellence / Contribution

- Targeted pest detection reduces unnecessary chemical use
- Modular design enables integration with automated spraying
- Computer vision approach supports scalable farm monitoring

User Feedback (Internal Evaluation)

- Automated detection reduces need for constant manual checks
- Targeted response concept viewed as environmentally friendly
- Mobile alerts identified as a critical future feature

Next Phase Enhancements

- Integration with automated neem spraying system
- Real-time alerts and pest location visualization

Commercialization , Target & Budget

Commercialization / Sustainability

- Smart monitoring system for small-scale bag mushroom farmers
- One system per cultivation hall
- Rail-mounted robot arm monitors all bag racks
- Reduces labor, crop loss, and manual inspection
- Scalable and reusable for long-term use

Target Farming Setup

- 1,000 – 1,500 mushroom bags
- Hall size: 20 ft × 10 ft × 5 ft (1000 square feet)
- Vertical rack arrangement

Estimated Cost (Per Hall)

- Robot Arm: LKR 180,000
- Rail + Sensors + Camera + Setup: LKR 55,000

Total: ~ LKR 235,000

Market , Uniqueness & Recovery

- Customer: Small-scale mushroom farmers (1–2 halls)
- Market: Growing bag-based mushroom farms in Sri Lanka
- Unique: Rail-mounted AI + IoT monitoring (low-cost)
- Selling Price: LKR 260,000 – 280,000 / hall
- Subscription: LKR 1,500 – 2,500 / month

Return of Investment for Farmers

- Farmers can recover the system cost in 12–18 months
- Based on:
 - Reduced crop loss
 - Lower labour cost
 - Improved yield consistency
 - Robot arm lifespan: ~ 5 years (long-term use)

Project Plan



Questions?



Thank You!